

COMMENTS: EM-BASED LIKELIHOOD INFERENCE FOR SOME LIFETIME DISTRIBUTIONS BASED ON LEFT TRUNCATED AND RIGHT CENSORED DATA AND ASSOCIATED MODEL DISCRIMINATION

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We thank the editor for inviting us to discuss this work of Professors Balakrishnan and Mitra. We comment firstly on their choice of EM as the route to the maximum likelihood estimates of parameters of their models, and secondly on their use of confidence intervals of Wald type. We pose three questions.

1. EM and direct numerical maximization

1.1. Direct numerical maximization instead of EM?

After describing the EM algorithm in Section 3, the authors show in detail, in Sections 4–7, how to apply it to their problem in the cases of lognormal, Weibull, gamma, and generalized gamma distributions. They use numerical maximization of the likelihood by Newton–Raphson in the lognormal, Weibull and gamma cases, but only ‘for comparison’. They describe such numerical maximization very briefly, and from their description it would appear to work smoothly and with relatively little effort, and to yield results very close in all cases to those supplied by EM. It seems that the implementation of numerical maximization of likelihood is straightforward, certainly far more so than their implementation of EM, which is really quite a *tour de force*.

So (somewhat provocatively) we pose this question: Knowing what they now know, would the authors choose to use EM in any problem similar to this, or instead to use numerical maximization via a general-purpose optimizer such as the one they did use, `maxNR`? Our impression is that, if one lacks the great skills and great determination of Balakrishnan and Mitra, one would be better off not spending one’s time on EM, but just setting out to use in the first place direct numerical maximization. Put differently, are there any clear advantages to EM in this application, or similar applications? There certainly seems to be a clear *disadvantage* here to EM: the huge amount of

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skilful (and potentially error-prone) work that is necessary. In their Section 3 the authors make a passing reference to the ‘many desirable and advantageous properties’ of EM as described by McLachlan and Krishnan (2008, no page reference given), but — perhaps for reasons of space — do not themselves mention what these properties are.

1.2. One-step Newton–Raphson maximization

A phrase which we noticed (in several places, in the context of the EM gradient algorithm) was ‘a one-step Newton-Raphson (NR) method.’ See e.g. Section 4.1 (method 2):

In this method, the function $Q(\boldsymbol{\theta}, \boldsymbol{\theta}^{(k)})$ is maximized in the M-step by a one-step Newton-Raphson (NR) method.

Is Q actually maximized (as a function of $\boldsymbol{\theta}$) in one step, or is it merely increased? Increasing the function Q is what is done in the M step of generalized EM, not maximization; see McLachlan and Krishnan (2008, p. 24). That is what distinguishes generalized EM from EM.

2. Confidence intervals

All the confidence intervals reported appear to be of Wald type, relying on the asymptotic normality of the MLEs. An alternative is the use of intervals based on the profile likelihood, which are computationally more intensive than those of Wald type, but less so than bootstrap intervals. Such likelihood-based intervals are relatively easy to implement once one has a likelihood evaluator available. Furthermore, Pawitan (2001, p. 42) writes that, if the Wald intervals differ from the likelihood-based intervals, the latter are preferable. McCullagh and Nelder (1989, p. 473) report that the chi-squared approximations on which the profile-likelihood intervals rest are often quite accurate even when the normal approximations for the distributions of the MLEs themselves are unsatisfactory.

Did the authors consider using such likelihood-based intervals?

As an experiment we took the (simulated) data in Table A.1 of Balakrishnan and Mitra (2014) and found intervals based on profile likelihood for the parameters μ and σ of the lognormal distribution there under discussion. For these data our (approximate) likelihood-based intervals are:

	μ	σ
90%	(3.4038, 3.6275)	(0.4257, 0.6303),
95%	(3.3822, 3.6513)	(0.4120, 0.6581).

Our intervals of Wald type are:

90%	(3.4036, 3.6224)	(0.4115, 0.6117),
95%	(3.3826, 3.6434)	(0.3923, 0.6309),

which agree (to three decimal places) with all the intervals other than bootstrap of Balakrishnan and Mitra (2014, Table 5, p. 412). It is clear that for μ it does not matter which of the two kinds of interval one uses; but for σ the profile likelihood is clearly asymmetric and the Wald interval therefore inappropriate. See our Figure 1.

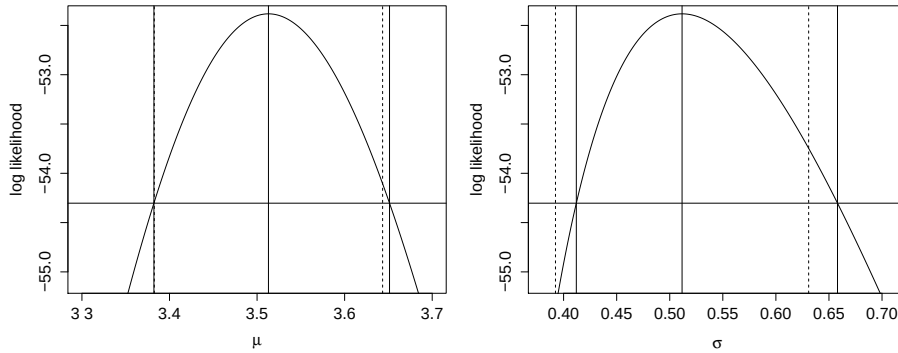


Figure 1: Data of Balakrishnan and Mitra (2014), profile log-likelihood functions for μ (left) and σ (right). The dashed vertical lines indicate the 95% confidence intervals of Wald type. The solid vertical lines indicate the MLEs (central solid lines) and the 95% confidence intervals based on profile likelihood. The horizontal line in each plot is at the level $\ell(\hat{\mu}, \hat{\sigma}) - \chi^2_{1,0.95}/2 = \ell(\hat{\mu}, \hat{\sigma}) - 1.92$, where ℓ is the log-likelihood.

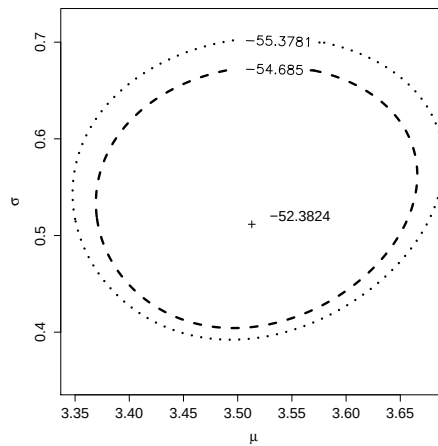


Figure 2: Data of Balakrishnan and Mitra (2014), likelihood-based 90% and 95% confidence regions for (μ, σ) . The MLE $(\hat{\mu}, \hat{\sigma})$ and associated log-likelihood are also marked on the plot.

But a more satisfactory way of representing the uncertainty in μ and σ is to find a likelihood-based joint confidence region for (μ, σ) . This is (for 90%) enclosed by the contour which lies log 10 units below the peak of the log-likelihood, and similarly log 20 units for 95%. The 90% and 95% confidence regions for (μ, σ) are displayed in our Figure 2.

3. Conclusion

In conclusion we suggest that, in problems such as these, maximum likelihood estimation should perhaps be done by direct numerical maximization of likelihood rather than by EM, and that confidence intervals or regions for parameters should be those based on the likelihood. At the very least this would greatly reduce the error-prone human effort involved in deriving and coding the EM algorithm and in then finding standard errors via the missing information principle; see Sections 4–7 of the paper under discussion for examples of the extensive work that is needed if one uses EM and the missing information principle.

References

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